THE SELF MAP GIVEN BY A RETRACTION IS HOMOTOPIC TO THE IDENTITY ON THE ASSOCIATED COSIMPLICIAL OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-push-outs-simplicial-object-w-section (source lines 5258-5269); the proof environment begins on the second line after it and is its own proof, closing at source line 5276.

Editorial note. The proof is a formal duality argument naming two other results; both are retained as context spans and the generator expands the context transitively (budget 800 lines for this card), so nothing is silently supplied.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Methods chapter, counted once: when f has a retraction, the degreewise self map on the pushout cosimplicial object is homotopic to the identity, so the object is homotopy equivalent to the constant one.

Allowed standard prerequisites (editorial).

- The pushout cosimplicial object. The cosimplicial object U with U_n the (n+1)-fold pushout of Y over X, built from f : X to Y when the category has pushouts (the author's example-push-outs-simplicial-object, retained as context).. It is the object of the statement (source lines 5260-5266).
- Retractions and the maps f s. A retraction s of f with s f = id, hence an endomorphism f s of Y appearing in each degree.. It produces the self map of the statement (source lines 5261-5266).
- The dual fibre-product lemma. For a morphism with a section, the degreewise self map on the fibre-product simplicial object is homotopic to the identity (the author's lemma-fibre-products-simplicial-object-w-section, retained as context).. It is the dual statement from which this one follows (source lines 5272-5274).
- The comparison of homotopies. The author's comparison lemma relating homotopies of simplicial and cosimplicial objects.. It transfers the dual statement to this setting (source lines 5274-5275).

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Example 0.1. Suppose that $X \to Y$ is a morphism of C such that all the pushouts $Y \amalg_X Y \amalg_X \ldots \amalg_X Y$ exist. Then we set U_n equal to the $(n+1)$-fold pushout, and we let $\varphi : [n] \to [m]$ correspond to the map

$$(y \text{ in } i\text{th component}) \mapsto (y \text{ in } \varphi(i)\text{th component})$$

on "coordinates". In other words, the map $U_1 = Y \amalg_X Y \to U_0 = Y$ is the identity on each component. The two maps $U_0 = Y \to U_1 = Y \amalg_X Y$ are the two coprojections.

Lemma 0.2. *Let $f : Y \to X$ be a morphism of a category $\mathcal{C}$ with fibre products. Assume f has a section s. Consider the simplicial object U constructed in Example 0.4 starting with f. The morphism $U \to U$ which in each degree is the self map $(s \circ f)^{n+1}$ of $Y \times_X \ldots \times_X Y$ given by $s \circ f$ on each factor is homotopic to the identity on U. In particular, U is homotopy equivalent to the constant simplicial object X.*

Proof. Set $g^0 = \mathrm{id}_Y$ and $g^1 = s \circ f$. We use the morphisms

$$\begin{aligned} Y \times_X \ldots \times_X Y \times \mathrm{Mor}([n], [1]) &\to Y \times_X \ldots \times_X Y \\ (y_0, \ldots, y_n) \times \alpha &\mapsto (g^{\alpha(0)}(y_0), \ldots, g^{\alpha(n)}(y_n)) \end{aligned}$$

where we use the functor of points point of view to define the maps. Another way to say this is to say that $h_{n,0} = \mathrm{id}$, $h_{n,n+1} = (s \circ f)^{n+1}$ and $h_{n,i} = \mathrm{id}_Y^{i+1} \times (s \circ f)^{n+1-i}$. We leave it to the reader to show that these satisfy the relations of Lemma 0.5. Hence they define the desired homotopy. See also Remark 0.6 which shows that we do not need to assume anything else on the category $\mathcal{C}$. □

Lemma 0.3. *Let $\mathcal{C}$ be a category. Suppose that U and V are two cosimplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of cosimplicial objects. Recall that U, V correspond to simplicial objects U', V' of $\mathcal{C}^{opp}$. Moreover a, b correspond to morphisms $a', b' : V' \to U'$. The following are equivalent*

1. *There exists a homotopy $h = \{h_{n,\alpha}\}$ from a to b as in Remark 0.9.*
2. *There exists a homotopy $h = \{h_{n,i}\}$ from a' to b' as in Remark 0.6.*

Thus a is homotopic to b as in Remark 0.9 if and only if a' is homotopic to b' as in Remark 0.6.

Proof. In case $\mathcal{C}$ has finite products, then $\mathcal{C}^{opp}$ has finite coproducts and we may use Definitions 0.8 and 0.7 instead of Remarks 0.9 and 0.6. In this case $h : U \to \mathrm{Hom}(\Delta[1], V)$ is the same as a morphism $h' : \mathrm{Hom}(\Delta[1], V)' \to U'$. Since products and coproducts get switched too, it is immediate that $(\mathrm{Hom}(\Delta[1], V))' = V' \times \Delta[1]$. Moreover, the "primed" version of the morphisms $e_0, e_1 : \mathrm{Hom}(\Delta[1], V) \to V$ are the morphisms $e_0, e_1 : V' \to \Delta[1] \times V$. Thus $e_0 \circ h = a$ translates into $h' \circ e_0 = a'$ and similarly $e_1 \circ h = b$ translates into $h' \circ e_1 = b'$. This proves the lemma in this case.

In the general case, one needs to translate the relations given by (0.8.1) into the relations given in Lemma 0.5. We omit the details.

The final assertion is formal from the equivalence of (1) and (2). □

Example 0.4. Suppose that $Y \to X$ is a morphism of $\mathcal{C}$ such that all the fibred products $Y \times_X Y \times_X \ldots \times_X Y$ exist. Then we set U_n equal to the $(n+1)$-fold fibre product, and we let $\varphi : [n] \to [m]$ correspond to the map (on "coordinates") $(y_0, \ldots, y_m) \mapsto (y_{\varphi(0)}, \ldots, y_{\varphi(n)})$. In other words, the map $U_0 = Y \to U_1 = Y \times_X Y$

is the diagonal map. The two maps $U_1 = Y \times_X Y \to U_0 = Y$ are the projection maps.

Lemma 0.5. *In the situation above, we have the following relations:*

(1) *We have* $h_{n,0} = b_n$ *and* $h_{n,n+1} = a_n$.
(2) *We have* $d^n_j \circ h_{n,i} = h_{n-1,i-1} \circ d^n_j$ *for* $i > j$.
(3) *We have* $d^n_j \circ h_{n,i} = h_{n-1,i} \circ d^n_j$ *for* $i \leq j$.
(4) *We have* $s^n_j \circ h_{n,i} = h_{n+1,i+1} \circ s^n_j$ *for* $i > j$.
(5) *We have* $s^n_j \circ h_{n,i} = h_{n+1,i} \circ s^n_j$ *for* $i \leq j$.

Conversely, given a system of maps $h_{n,i}$ *satisfying the properties listed above, then these define a morphism* h *which is a homotopy from* a *to* b.

Proof. Omitted. You can prove the last statement using the fact, see Lemma 0.10 that to give a morphism of simplicial objects is the same as giving a sequence of morphisms h_n commuting with all d^n_j and s^n_j. □

Remark 0.6. Let $\mathcal{C}$ be any category (no assumptions whatsoever). Let U and V be simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of simplicial objects of $\mathcal{C}$. A *homotopy from* a *to* b is given by morphisms[1] $h_{n,i} : U_n \to V_n$, for $n \geq 0$, $i = 0, \ldots, n+1$ satisfying the relations of Lemma 0.5. As in Definition 0.7 we say the morphisms a and b are *homotopic* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}. Clearly, if $F : \mathcal{C} \to \mathcal{C}'$ is any functor and $\{h_{n,i}\}$ is a homotopy from a to b, then $\{F(h_{n,i})\}$ is a homotopy from $F(a)$ to $F(b)$. Similarly, if a and b are homotopic, then $F(a)$ and $F(b)$ are homotopic. Since the lemma says that the newer notion is the same as the old one in case finite coproduct exist, we deduce in particular that functors preserve the original notion whenever both categories have finite coproducts.

Definition 0.7. Let $\mathcal{C}$ be a category having finite coproducts. Suppose that U and V are two simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be two morphisms.

(1) We say a morphism
$$h : U \times \Delta[1] \longrightarrow V$$
is a *homotopy from* a *to* b if $a = h \circ e_0$ and $b = h \circ e_1$.
(2) We say the morphisms a and b are *homotopic* or are *in the same homotopy class* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}.

Definition 0.8. Let $\mathcal{C}$ be a category having finite products. Let U and V be two cosimplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be two morphisms of cosimplicial objects of $\mathcal{C}$.

(1) We say a morphism
$$h : U \longrightarrow \mathrm{Hom}(\Delta[1], V)$$
such that $a = e_0 \circ h$ and $b = e_1 \circ h$ is a *homotopy from* a *to* b.

[1] In the literature, often the maps $h_{n+1,i} \circ s_i : U_n \to V_{n+1}$ are used instead of the maps $h_{n,i}$. Of course the relations these maps satisfy are different from the ones in Lemma 0.5.

(2) We say a and b are *homotopic* or are *in the same homotopy class* if there exists a sequence $a = a_0, a_1, \ldots, a_n = b$ of morphisms from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_i to a_{i-1} or there exists a homotopy from a_{i-1} to a_i.

$$h_{m,\alpha} \circ U(f) = V(f) \circ h_{n,\alpha \circ f} \tag{0.8.1}$$

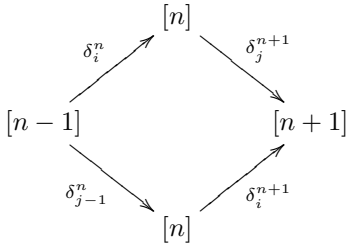
Remark 0.9. Let $\mathcal{C}$ be any category (no assumptions whatsoever). Let U and V be cosimplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of cosimplicial objects of $\mathcal{C}$. A *homotopy from a to b* is given by morphisms $h_{n,\alpha} : U_n \to V_n$, for $n \geq 0$, $\alpha \in \Delta[1]_n$ satisfying (0.8.1) for all morphisms f of Δ and such that $a_n = h_{n,0:[n]\to[1]}$ and $b_n = h_{n,1:[n]\to[1]}$ for all $n \geq 0$. As in Definition 0.8 we say the morphisms a and b are *homotopic* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}. Clearly, if $F : \mathcal{C} \to \mathcal{C}'$ is any functor and $\{h_{n,i}\}$ is a homotopy from a to b, then $\{F(h_{n,i})\}$ is a homotopy from $F(a)$ to $F(b)$. Similarly, if a and b are homotopic, then $F(a)$ and $F(b)$ are homotopic. This new notion is the same as the old one in case finite products exist. We deduce in particular that functors preserve the original notion whenever both categories have finite products.

Lemma 0.10. *The category Δ is the universal category with objects $[n]$, $n \geq 0$ and morphisms δ^n_j and σ^n_j such that (a) every morphism is a composition of these morphisms, (b) the relations listed in Lemma 0.11 are satisfied, and (c) any relation among the morphisms is a consequence of those relations.*

Proof. Omitted. □

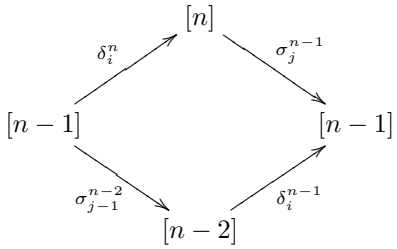
Lemma 0.11. *The morphisms δ^n_j and σ^n_j satisfy the following relations.*

(1) *If $0 \leq i < j \leq n+1$, then $\delta^{n+1}_j \circ \delta^n_i = \delta^{n+1}_i \circ \delta^n_{j-1}$. In other words the diagram*



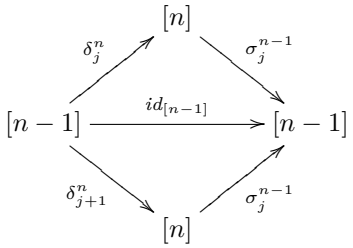
commutes.

(2) *If $0 \leq i < j \leq n-1$, then $\sigma^{n-1}_j \circ \delta^n_i = \delta^{n-1}_i \circ \sigma^{n-2}_{j-1}$. In other words the diagram*



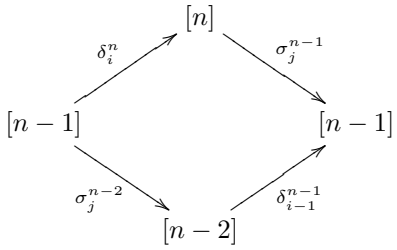
commutes.

(3) *If* $0 \leq j \leq n-1$, *then* $\sigma_j^{n-1} \circ \delta_j^n = id_{[n-1]}$ *and* $\sigma_j^{n-1} \circ \delta_{j+1}^n = id_{[n-1]}$. *In other words the diagram*



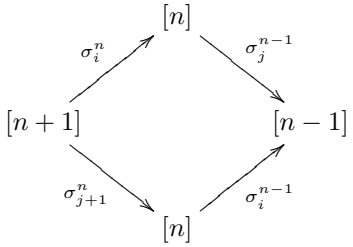
commutes.

(4) *If* $0 < j+1 < i \leq n$, *then* $\sigma_j^{n-1} \circ \delta_i^n = \delta_{i-1}^{n-1} \circ \sigma_j^{n-2}$. *In other words the diagram*



commutes.

(5) *If* $0 \leq i \leq j \leq n-1$, *then* $\sigma_j^{n-1} \circ \sigma_i^n = \sigma_i^{n-1} \circ \sigma_{j+1}^n$. *In other words the diagram*



commutes.

Proof. Omitted. □

Lemma 0.12. *Let* $f : X \to Y$ *be a morphism of a category* $\mathcal{C}$ *with pushouts. Assume there is a morphism* $s : Y \to X$ *with* $s \circ f = id_X$. *Consider the cosimplicial object* U *constructed in Example 0.1 starting with* f. *The morphism* $U \to U$ *which in each degree is the self map of* $Y \amalg_X \ldots \amalg_X Y$ *given by* $f \circ s$ *on each factor is homotopic to the identity on* U. *In particular,* U *is homotopy equivalent to the constant cosimplicial object* X.

Proof. This lemma is dual to Lemma 0.2. Hence this lemma follows on applying Lemma 0.3. □